Neerav Chaudhary

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SUMMARY

AI/ML Engineer and Data Scientist with hands-on industry and research experience developing LLM-powered tools, predictive models, and computer vision systems for healthcare. Improved detection accuracy and reduced false positives with custom churn-prediction solutions, and published deep learning research on brain atrophy quantification. Aims to use applied AI skills to support impactful, data-driven outcomes in healthcare environments.

EDUCATION

San Jose State University **Aug 2024 - May 2026**

M.S., Industrial / Engineering Management (GPA: 3.81)

- **Coursework:** Math Foundations for Data Science, Advanced Systems Engineering, Lean Six Sigma, Advanced Operations Control

Manipal University Jaipur **Aug 2018 - Jun 2022**

Bachelor's, Computer Engineering (GPA: 8.8)

PROFESSIONAL EXPERIENCE

Deep Strat | *Artificial Intelligence Research Intern* **Aug 2023 - Nov 2023**

- Built churn and loan-default prediction models (Random Forest, CNN), designing a custom loss function that jointly optimized false-positive rate and profit to maximize detection reliability, reducing false positives by 25% while maintaining stable AUC
- Prototyped an LLM-powered ticket assistant that performed sentiment and intent analysis on support emails and chat logs, auto-classified and routed issues to departments via vectorized data retrieval, and generated SOP-grounded response drafts to standardize resolution quality
- Authored a 50-page technical reference spanning the full ML/AI stack - from regression fundamentals to LLM architectures, failure modes, and bias propagation - used to diagnose specific causes of model compliance and ethical failures for stakeholders
- Contributed technical analysis and stakeholder discussions that informed a jointly authored conference paper on AI safety and responsible deployment practices, presented at Nullcon

Manipal University Jaipur | *AI Researcher, Medical Imaging* **Aug 2022 - Dec 2022**

- Designed and evaluated CNN architectures for dementia severity classification on the ADNI longitudinal MRI dataset, testing depth-width trade-offs and combining models in an ensemble framework that improved accuracy from 78% to 85%
- Implemented an OpenCV-based method to quantify neural atrophy from MRI scans, automating extent-of-damage estimation to reduce the time required for manual assessment of disease progression (published in IEEE ICRITO 2022)
- Built image-processing workflows to segment affected brain regions and extract structural features, linking imaging outputs to clinically meaningful indicators of atrophy severity across patient cohorts aimed at helping clinicians prioritize patients for further evaluation,

PUBLICATIONS

- N. Chaudhary, S. S. Verma.Analyzing Brain Atrophy in Dementia Using Computer Vision..*10th Intl. Conf. on Reliability, Infocom Technologies and Optimization (ICRITO), IEEE, 2022*

PROJECTS

Refract-AI Research and Statistical Analysis Platform **Jan 2026 - May 2026**

- Integrated RAG-based literature synthesis with automated statistical analysis in one research environment, cutting end-to-end workflow time by 45% (10 hrs to under 6) by eliminating tool-switching across PDF readers, reference managers, and statistical software (React, FastAPI, PostgreSQL, ChromaDB)
- Automated EDA, leakage-prone variable removal (Spearman $r \geq 0.97$), correlated-pair screening for dimensionality reduction, and staged baseline-to-random-forest model comparison, validated by reproducing a previously validated Battery RUL prediction (RMSE 25.6 cycles)
- Engineered a structured comparison layer converting multi-paper review (10+ research papers per session) into reusable artifacts such as cross-paper difference matrices, topic coverage mapping, and research gap identification with AI-augmented interpretations classified by evidence provenance using session-scoped retrieval to prevent ungrounded claims

Predictive Maintenance of EV Batteries: Preventing Failures & Minimizing Waste **Sep 2024 - Dec 2024**

- Developed a predictive maintenance pipeline to estimate the remaining useful life (RUL) of lithium-ion batteries using historical degradation data. Engineered features from impedance and capacity-fade trends and trained Random Forest regression models.
- Achieved R^2 0.993 and RMSE 25.61 cycles predicting battery remaining useful life, with Discharge_Time carrying ~85% feature importance after removing leakage-prone variables through correlation-based audit

SKILLS

- **Languages & Data:** Python, C++, R, SQL, Git
- **ML & Deep Learning:** Regression, Classification, Ensemble Methods, Random Forest, CNN, Feature Engineering, Model Evaluation, Predictive Modeling, Forecasting, OpenCV
- **Statistics & Analysis:** Hypothesis Testing, Chi-Square, Statistical Analysis, EDA, Dimensionality Reduction, Lean Six Sigma
- **NLP & LLM Systems:** Large Language Models, Retrieval-Augmented Generation, Prompt Engineering, Sentiment Analysis, Vector Databases, LangChain, Hugging Face, FAISS
- **Tools & Frameworks:** TensorFlow, PyTorch, scikit-learn (sklearn), Matplotlib, Seaborn, Plotly, ggplot2, PostgreSQL, ChromaDB